# Volatility Risk Premium and Non-Directional Option Strategies on the S&P 500 Index: Evidence from 2010–2023 SPX Option Chains

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**Extended abstract**

**Question.** This paper tests whether the prices of European options on the S&P 500 index contain a volatility risk premium (VRP) and whether non-directional premium-selling strategies (iron condors, credit spreads, strangles and related structures) capture it net of costs.

**Data and method.** End-of-day option chains of SPX options (European-style, cash-settled index options listed on Cboe; source OptionsDX) from January 2010 to December 2023: 3,499 sessions and 47,699 date–expiry pairs. Implied volatility is obtained from prices with the Black (1976) model on the forward implied by put-call parity; model-free implied variance follows the Cboe formula; realized variance is measured up to the expiry of the same option. The real-world forecast is a GJR-GARCH(1,1) with Student-$t$ innovations estimated with no look-ahead. The backtests cover 434 configurations of ten families, for 1,310,509 simulated trades, at mid and at bid-ask prices.

**Results.** (i) The 30-day model-free volatility exceeds realized volatility by +3.45 points on average ($t$ 6.48); at-the-money volatility exceeds it by +0.72 points ($t$ 1.36). The premium sits in the wings, not at the center of the surface. (ii) The at-the-money premium is significant in 2010–2014 (+1.82 points, $t$ 4.02) and zero in 2015–2019 and 2020–2023. (iii) The 30-day at-the-money implied volatility is an unbiased forecast of realized volatility (slope 0.908, joint test $p = 0.409$) with a smaller error than GARCH and historical volatility. (iv) 0.10-delta, 30-day puts returned to their buyers a fraction of 0.44 of the premium paid. (v) The median Sharpe ratio across configurations is +0.39 at mid prices and $-0.40$ at bid-ask prices; the share with a significantly positive alpha against the index falls from 16.4% to 2.5%, the value expected by chance alone. (vi) Positive results are concentrated in structures that sell puts and carry positive equity beta; iron condors and straddles have a zero or negative net result; early-exit rules lower returns because they pay a second spread. (vii) Suspending entries when realized volatility exceeds 25% does not improve the Sharpe ratio (median from $-0.30$ to $-0.34$) and does not reduce the worst month: extreme losses originate in calm phases, and delta-selected strikes already widen with implied volatility (from 3.8% to 7.5% of the forward between the low and the high tercile). (viii) Returns are far from normal: for the 0.16-delta, 45-day short put per-trade skewness is $-9.5$ and excess kurtosis 108, against $-1.1$ and 4.8 for the index over the same horizon, and normality of the monthly series is rejected in 92% of configurations. The Sharpe ratio therefore overstates short-volatility structures, and tail risk has to be measured directly.

| | Mid prices | Bid-ask prices |
|---|---|---|
| Median Sharpe ratio, 434 configurations | +0.39 | −0.40 |
| Share with positive Sharpe ratio (%) | 64 | 38 |
| Share with positive alpha, $t > 2$ (%) | 16.4 | 2.5 |
| Short put Δ0.16, 45 days: Sharpe | +0.99 | +0.93 |
| Put credit spread Δ0.20/0.05, 30 days: Sharpe | +0.98 | +0.83 |
| Iron condor Δ0.16/0.05, 30 days: Sharpe | +0.43 | +0.14 |

**Conclusion.** The premium exists as compensation for tail risk and is consistent with a market that is informationally efficient at the center of the surface. The bid-ask spread, the embedded equity exposure and ex-post parameter selection explain almost all of the positive backtest results. The hypothesis that systematic capture of the premium tends, over the long run and net of costs, to a zero result is not rejected by the data.

**Keywords:** volatility risk premium, index options, iron condor, put credit spread, GJR-GARCH, transaction costs, multiple testing.

Authors: Giordano Margotti, Alex Crescentini, Alessandro Celani, Giulio Sciacchitano (*finance-thinking* working group).
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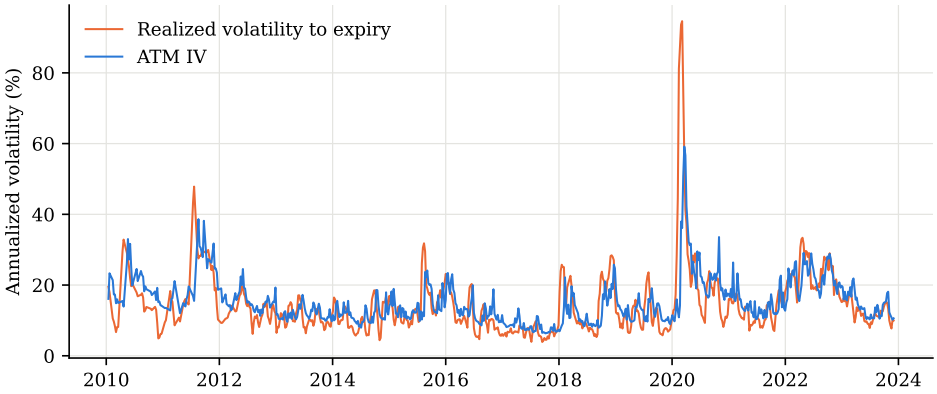



**Figure 1.** At-the-money IV on the expiry closest to 30 days and realized volatility to that expiry.

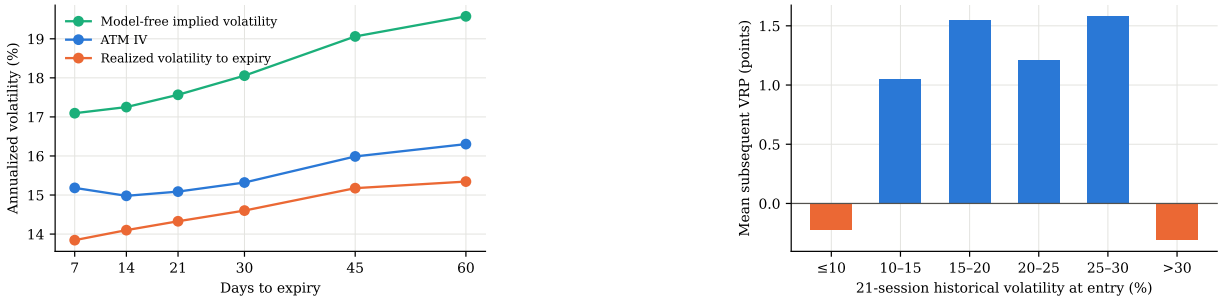



**Figure 2.** Left: means by horizon of model-free IV, at-the-money IV and realized volatility. Right: mean subsequent at-the-money premium by level of historical volatility at entry.

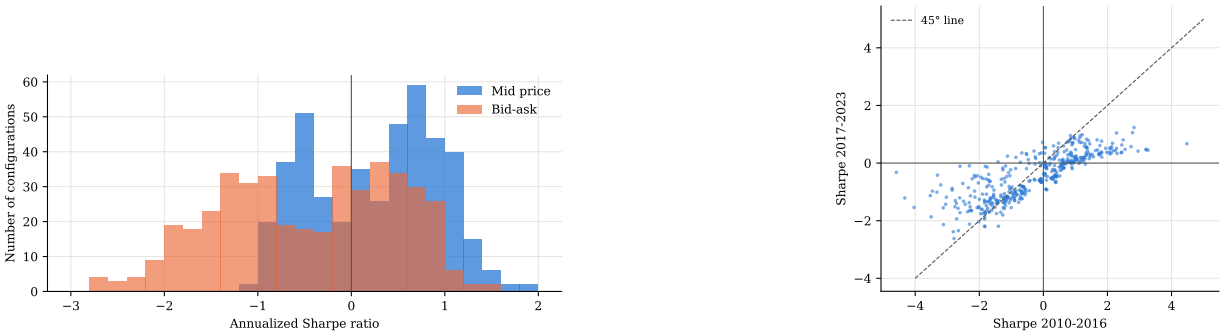



**Figure 3.** Left: distribution of the Sharpe ratio across the 434 configurations at mid and bid-ask prices. Right: Sharpe ratio in 2010–2016 and in 2017–2023 for credit configurations.

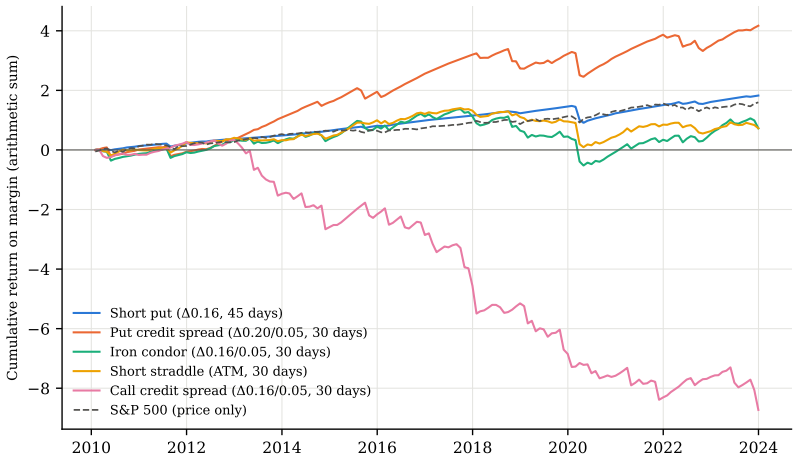



**Figure 4.** Cumulative return on margin at bid-ask prices for five configurations and for the index.